**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 085 · Insertion Sort

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. When does ascending bubble sort swap two adjacent items?

Q2. How does the OS support multitasking?
A. Shares CPU time
B. Adds a CPU per app
C. Closes every other app
D. Gives each app all RAM

Q3. “Open source means there are never any licence conditions or costs.” Correct this. Explain why.

Q4. A school uses biased software to choose pupils. A business wants to share licensed software. Correct both claims. Explain why. (a) Every legal use is ethical. (b) Law depends only on what is popular.

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# I Do · Read new knowledge

## Key vocabulary

Insertion sort — Builds a sorted section by inserting one value at a time.

Key — Value currently being inserted.

Sorted portion — Part of the array already in ascending or descending order.

Shift — Move a value one position to make room.

Compare — Test two values to decide their order.

Ascending — Ordered from smallest to largest.

## New learning

Sorted prefix: Insertion sort begins with the first item treated as a sorted portion.

Choose a key: Take the next unsorted value and compare it with values already in the sorted portion.

Shift larger entries: Move larger sorted values one position right to make a gap; do not lose the saved key.

Insert and repeat: Place the key into the gap and extend the sorted portion until the array is fully ordered.

## Worked example

Trace insertion sort on [4,1,3] and give the array after each insertion. [4 marks]

1. Initially [4] is treated as sorted [1].

2. Insert key 1 before 4, shifting 4 right: [1,4,3] [1].

3. Next key 3 is compared with 4 and 4 shifts right [1].

4. Insert 3 after 1 to form [1,3,4] [1].

# We Do · Practise with guidance

### Guided 1

Apply insertion sort to [5,2,4]. Show the array after inserting 2 and after inserting 4. [4 marks]

Worked method: Treat 5 as sorted [1]; shift 5 right and insert 2: [2,5,4] [1]. Move 5 right [1]; insert 4 to get [2,4,5] [1].

### Guided 2

Explain what happens when inserting 7 into sorted prefix [2,5,8]. [3 marks]

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Worked method: Compare 7 with 8 [1], shift 8 one position right [1], place 7 after 5 and before 8 [1].

### Guided 3

Explain one difference between insertion and bubble sorts in how they reorder data. [4 marks]

Worked method: Insertion sort takes a key [1] and inserts it into a growing sorted section [1]. Bubble sort compares adjacent pairs [1] and swaps out-of-order neighbours over passes [1].

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# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Code First · Predict → Write → Test → Improve

Fill the missing swap assignment in the first insertion step.

```
array sizes = [4,1,3]
key = sizes[1]
sizes[1] = sizes[0]
... = key
print(sizes[0])
print(sizes[1])
```

write

### Question 1

State what is meant by the key in insertion sort. [2 marks] Source: Original OCR J277-style question; not an official past-paper original.

### Question 2

Show insertion sort on [6,3,5] after each insertion. [4 marks] Source: Original OCR J277-style question; not an official past-paper original.

### Question 3

Explain how insertion sort orders [7,2,5,1] by maintaining a sorted section and shifting values. [6 marks] Source: Original OCR J277-style question; not an official past-paper original.

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## Exit Ticket · Answer before checking

Insertion sort inserts each new key into the correct position of a growing sorted part.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. When the left item is greater than the right item.

Q2. A — Shares CPU time

Q3. It permits access to source code and modification or redistribution under its conditions.

Q4. (a) Ethics concerns fairness, responsibility and what is right. (b) Law sets what is allowed or banned. Consider effects on people: a legal action may still be unfair or intrusive.

## I Do / We Do pages · Guided practice

Q1. Treat 5 as sorted [1]; shift 5 right and insert 2: [2,5,4] [1]. Move 5 right [1]; insert 4 to get [2,4,5] [1].

Q2. Compare 7 with 8 [1], shift 8 one position right [1], place 7 after 5 and before 8 [1].

Q3. Insertion sort takes a key [1] and inserts it into a growing sorted section [1]. Bubble sort compares adjacent pairs [1] and swaps out-of-order neighbours over passes [1].

## You Do pages · Code First

```
array sizes = [4,1,3]
key = sizes[1]
sizes[1] = sizes[0]
sizes[0] = key
print(sizes[0])
print(sizes[1])
```

Expected result: 1 4

## You Do pages · Independent questions

Q1. The current unsorted element being inserted [1] into its correct place among the already-sorted elements [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Q2. Initially 6 is sorted [1]. Shift 6 and insert 3 to produce [3,6,5] [1]. Shift 6 again [1] and insert 5 to get [3,5,6] [1].

Q3. Start with [7] as sorted [1]. Insert 2 before 7 → [2,7,5,1] [1]. Insert 5 before 7 → [2,5,7,1] [1]. For key 1 shift 7 then 5 then 2 right [2]. Insert 1 first → [1,2,5,7] [1].

### Exit Ticket · Model answer

True

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**